Visualization Design Guidelines and Principles

Arpit Narechania

Agenda

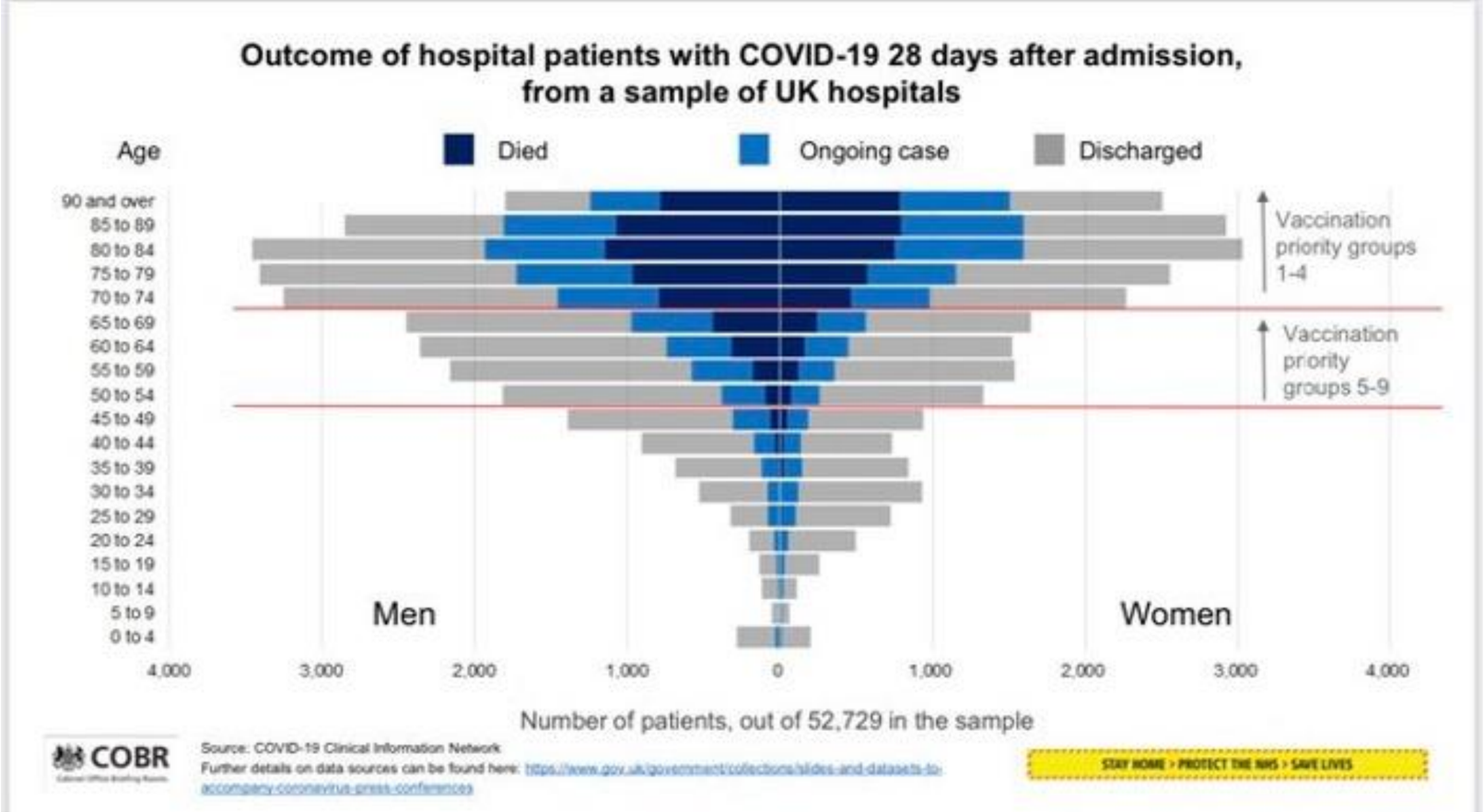
- Design (and redesign) as a process to help us think about the problem we are designing against
- Design studies
- Design guidelines and principles – a preview

Design

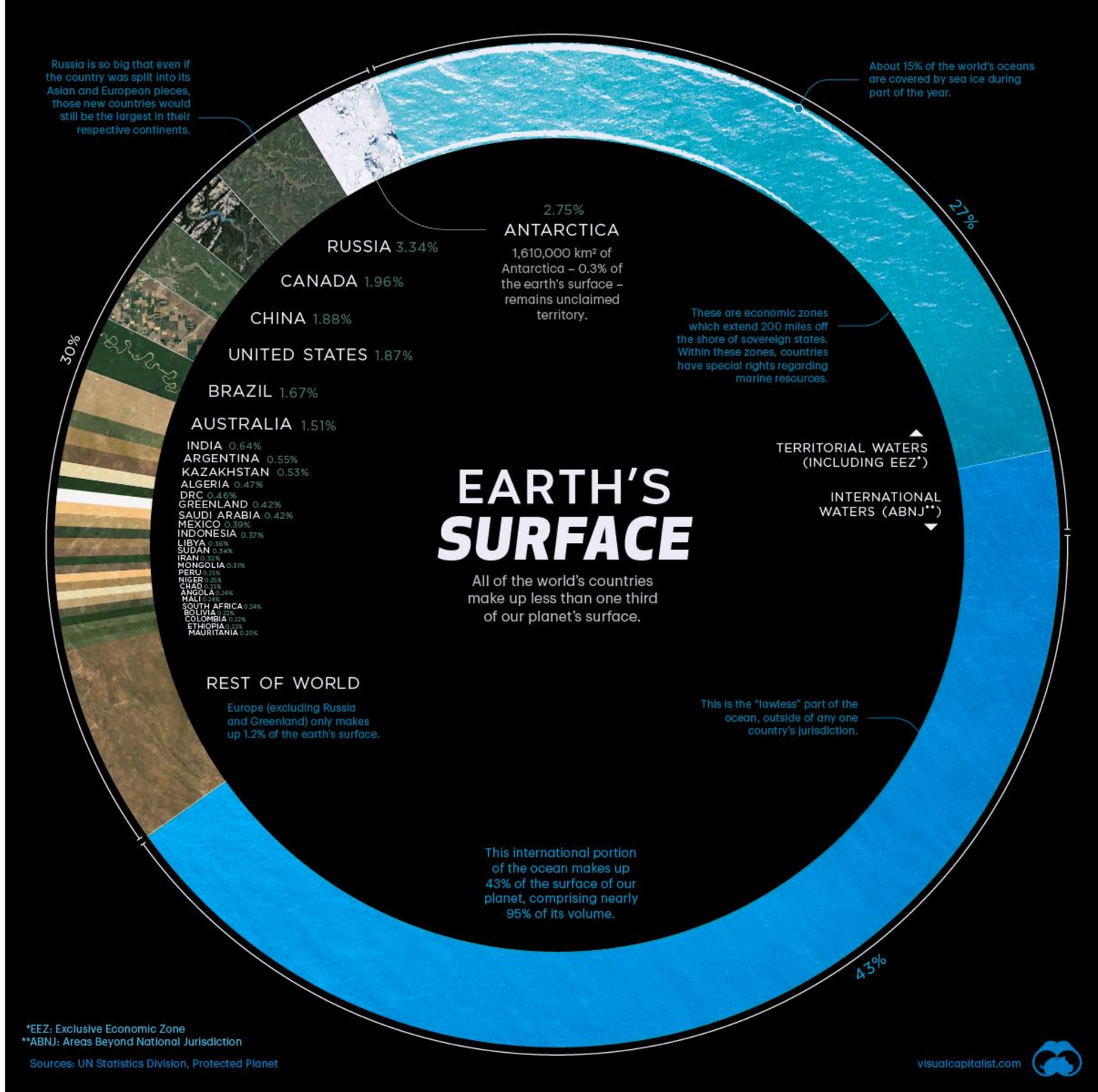
- Thinking about higher-level reasons or purposes to design brings us back to considering the use/need/purpose of the visualization

We can think about tasks

- Which type of tasks work well?
- Which not so well?



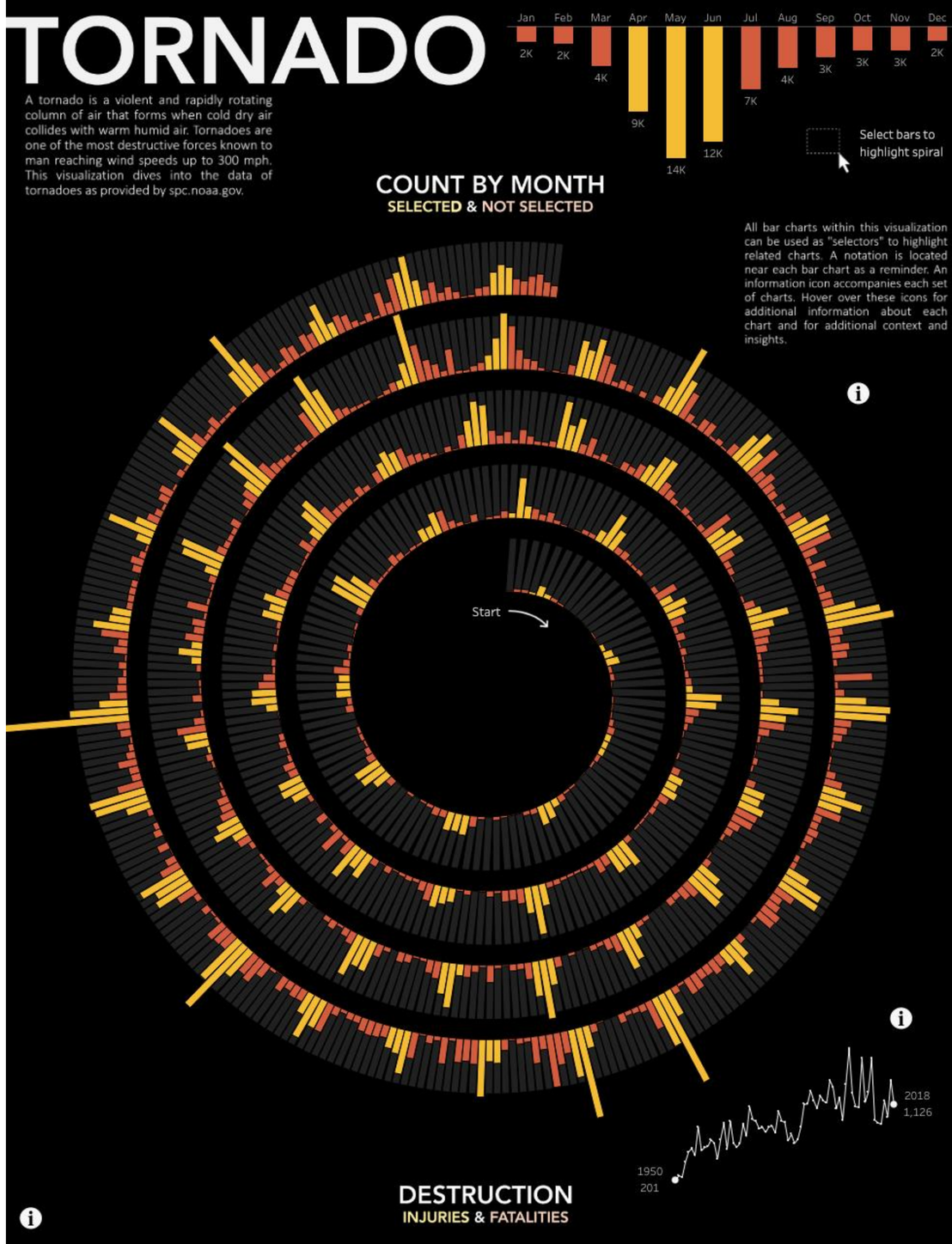
Or also



https://www.visualcapitalist.com/countries-by-share-of-earths-surface/

Or interactivity mixed in

https://public.tableau.com/app/profile/kevin.flerlage/viz/TORNADO/Tornado?publish=yes



More than graphical integrity?

- Many of these designs are not “optimal” per Tufte, Few, etc.
  - Some would argue that a “generalizable optimal visualization design” doesn’t exist
  - Always relative?
  - Thoughts?

- So we should consider a broader context for designing visualizations

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Research Papers

On the role of design in information visualization

Andrew Vande Moere[1] and Helen Purchase[2]

Abstract

Every information visualization developer is engaged in a complex process of design – balancing the three requirements of utility, soundness and attractiveness within given constraints. This paper investigates the oft-overlooked requirement of attractiveness (or 'aesthetics') in visualization, and, in doing so, discusses a wider role for the design process which emphasizes the benefits of disseminating design rationale. In presenting a model of three potential roles for design in information visualization, we suggest that the field would benefit from encompassing a broader scope that includes visualizations produced as part of commercial practice or artistic exploration. We conclude with a discussion on the practical consequences to the academic community of adopting our model – consequences that will require a paradigm shift in the way we value, teach and conduct information visualization research.

Keywords

design, visualization, aesthetics, case studies, design studies

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Notable quotes

- Design is "the **conception and realization of new things**" which has "its own distinct things to know, ways of knowing them, and ways of finding out about them." [Cross '82]

- Designer is someone "who **devises courses of action aimed at changing existing situations into preferred ones**". [Simon '96]

- Designing is "a pattern of behaviour employed in **inventing things of value which do not yet exist**". [Gregory '66]

- Design consists of various **goal-oriented, constrained, decision-making, exploration and learning activities**, which all operate within a given context, and which depend on the designer's perception of that context. [Gero '90]

"Every information visualization developer is **engaged in a complex process of design** –

balancing the three requirements of **utility**, **soundness** and **attractiveness**

within given constraints."

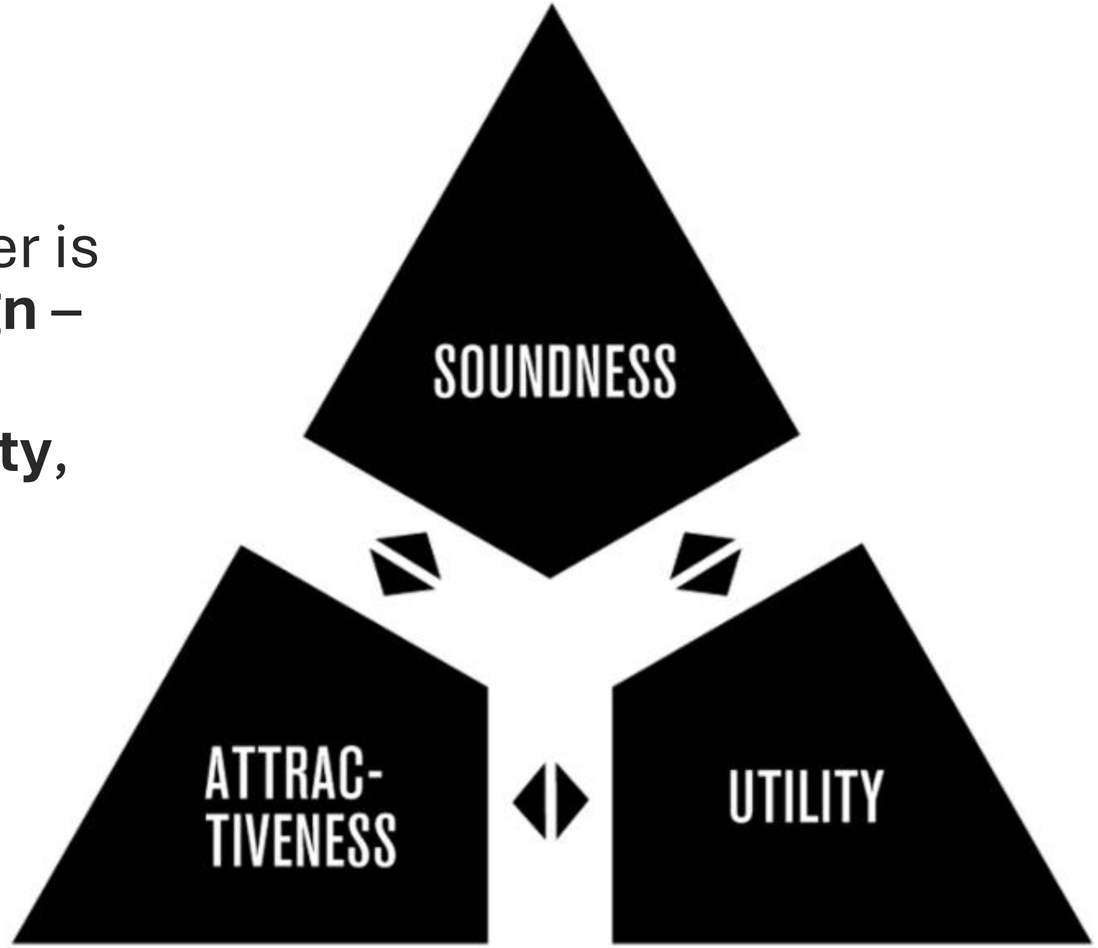


Figure 5. The Vitruvius triangle, based on typical architectural design requirements.

<u>Utility</u> - Corresponds to the **classic notions of functionality, usability, usefulness and other quantitative performance measures**. In visualization evaluation studies, these aspects are generally defined in terms of **effectiveness** (the accuracy and completeness with which users achieve specific tasks) and **efficiency** (the amount of resources expended in relation to the effectiveness criterion).

<u>Soundness</u> - Concerned with **reliability** and **robustness**. In the context of information visualization, soundness relates to the quality of the visualization presentation algorithm: Does it crash? Does it do what it is intended to do? How efficient is the code? Soundness also refers to the extent of the algorithm's wider use, outside of its initial presentation: Can it represent a wide range of data sets? Can it be used with different output media? Can it withstand variation in data complexity (e.g. size, dimensionality, time variance) or data inaccuracies?

<u>Attractiveness</u> – a.k.a. '**aesthetics**': the **appeal** or **beauty** of a given solution. Aesthetics does not limit itself to the visual form, but also includes **fuzzy aspects such as originality, innovation and novelty, and other subjective factors** comprising the user experience.

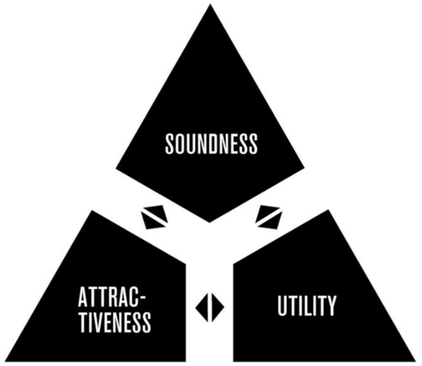
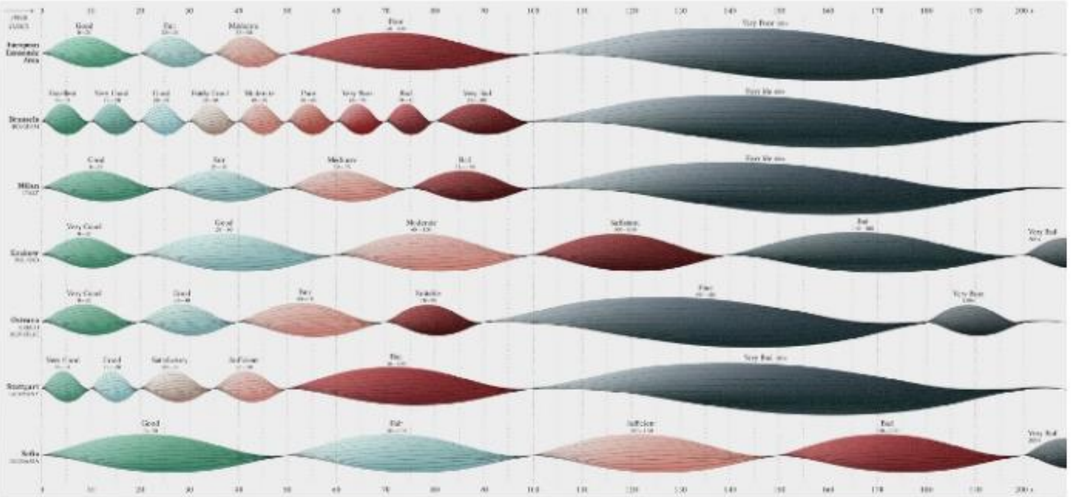


Figure 5. The Vitruvius triangle, based on typical architectural design requirements.

Elegance also plays a role

- Aesthetically Nourishing
- Stylistically coherent
- Refined
- Accessible
- Functionally delightful
- Noticeable when absent



"A Noxious Problem" by Federica Fragapane // CC BY-SA 3.0

What Makes A Data Visualisation Elegant?

Data Visualization Society members share their views on this elusive design concept.

On a dedicated channel, #dvs-topics-in-data-viz, in the Data Visualization Society Slack, our members discuss questions and issues pertinent to the field of data visualization. Discussion topics rotate every two weeks, and while subjects vary, each one challenges our members to think deeply and holistically about questions that affect the field of data visualization. At the end of each discussion, the moderator recaps some of the insights and observations in a post on Nightingale. You can find all of the other discussions here.

Assertions

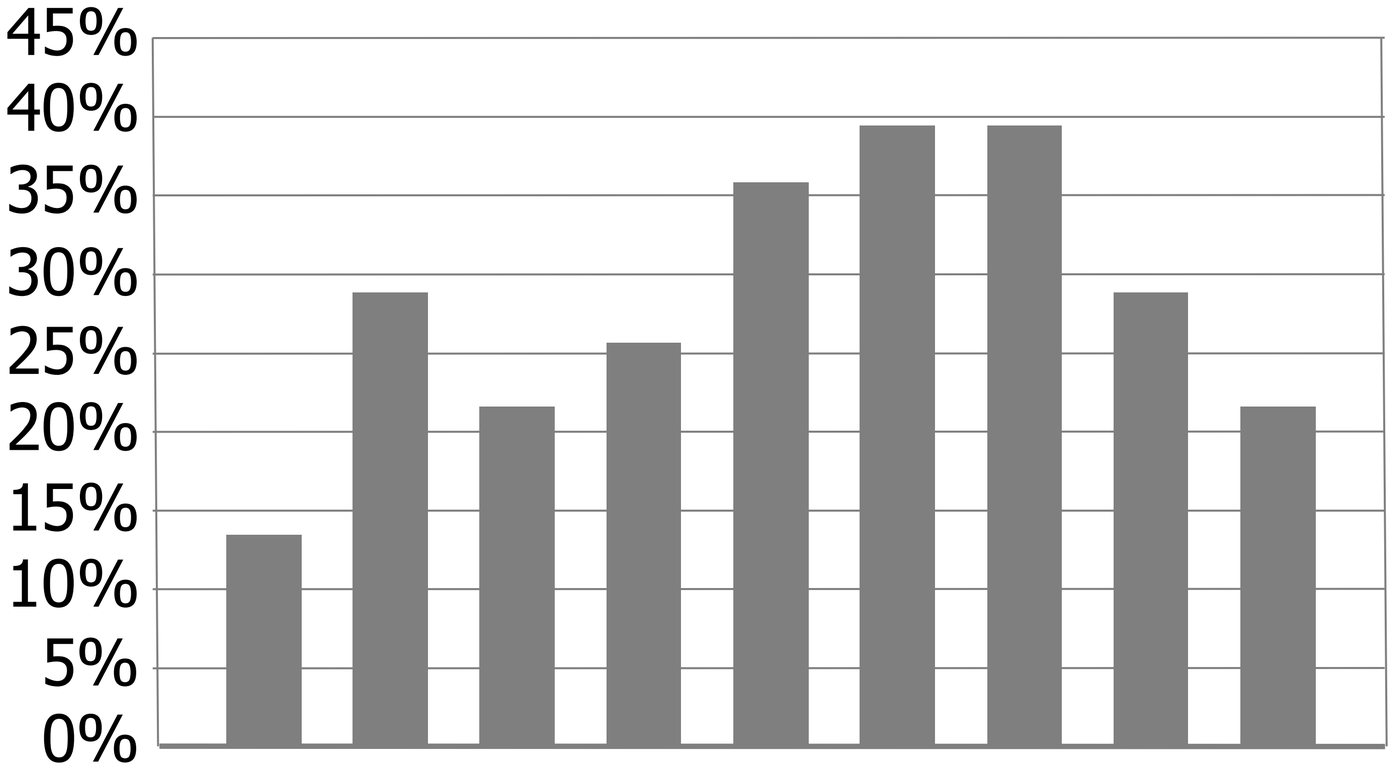
- Visualization design is **not an "add on"**
  - Does not get "added at the end"
  - Is not different from functionality or utility of the vis, but rather a part of it
- Visualization design is **not a black box**
  - Not some mystic activity, but grounded in rationale
- Visualization design is **not purely art**
  - Still has some functional purpose
- Visualization design is **not purely engineering**
  - "getting it to work" is only part of the challenge
- Visualization design **can be disseminated**
  - Should articulate and support design decisions made
- Visualization design **can be learned**
  - Practicing and learning can improve how one designs
- Visualization design **is part of a broader community**
  - HCI, UX, UI, information design, graphic design, …

Let’s try one

- Re-design this vis
  - Consider elegance, soundness, utility

- Take some time, think about how to do this, then sketch it

Bar chart



Here’s another one

- Re-design this vis
  - Consider task
  - Consider elegance, soundness, utility
  - Consider balance between these

- Take some time, think about how to do this, then sketch it

Time to Close Deal
Goal = 90 days
140.0
120.0
100.0
80.0
60.0
40.0
20.0
0.0
88.2
82.2
76.3
71.4
47.8
88.7
76.1
81.0
71.4
88.4
58.6
120.2
79.9
83.5
69.4
73.8
53.9
98.0
80.8
85.1
74.3
94.4
46.4
64.8
Jan-19
Feb-19
Mar-19
Apr-19
May-19
Jun-19
Jul-19
Aug-19
Sep-19
Oct-19
Nov-19
Dec-19
Direct Sales
Indirect Sales

Design Studies

Design Study Methodology

"A design study is a project in which visualization researchers

- analyze a specific real-world problem faced by domain experts,
- design a visualization system that supports solving this problem,
- validate the design, and
- reflect about lessons learned in order to

refine visualization design guidelines."

Design Study Methodology: Reflections from the Trenches and the Stacks

Michael Sedlmair, *Member, IEEE*, Miriah Meyer, *Member, IEEE*, and Tamara Munzner, *Member, IEEE*

Abstract—Design studies are an increasingly popular form of problem-driven visualization research, yet there is little guidance available about how to do them effectively. In this paper we reflect on our combined experience of conducting twenty-one design studies, as well as reading and reviewing many more, and on an extensive literature review of other field work methods and methodologies. Based on this foundation we provide definitions, propose a methodological framework, and provide practical guidance for conducting design studies. We define a design study as a project in which visualization researchers analyze a specific real-world problem faced by domain experts, design a visualization system that supports solving this problem, validate the design, and reflect about lessons learned in order to refine visualization design guidelines. We characterize two axes—a *task clarity axis* from fuzzy to crisp and an *information location axis* from the domain expert's head to the computer—and use these axes to reason about design study contributions, their suitability, and uniqueness from other approaches. The proposed methodological framework consists of 9 stages: *learn*, *winnow*, *cast*, *discover*, *design*, *implement*, *deploy*, *reflect*, and *write*. For each stage we provide practical guidance and outline potential pitfalls. We also conducted an extensive literature survey of related methodological approaches that involve a significant amount of qualitative field work, and compare design study methodology to that of ethnography, grounded theory, and action research.

Index Terms—Design study, methodology, visualization, framework.

1 Introduction

Over the last decade design studies have become an increasingly popular approach for conducting problem-driven visualization research. Design study papers are explicitly welcomed at several visualization venues as a way to explore the choices made when applying visualization techniques to a particular application area [55], and many exemplary design studies now exist [17, 34, 35, 56, 94]. A careful reading of these papers reveals multiple steps in the process of conducting a design study, including analyzing the problem, abstracting data and tasks, designing and implementing a visualization solution, evaluating the solution with real users, and writing up the findings.

And yet there is a lack of specific guidance in the visualization literature that describes holistic methodological approaches for conducting design studies—currently only three paragraphs exist [49, 55]. The relevant literature instead focuses on methods for designing [1, 42, 66, 79, 82, 90, 91] and evaluating [13, 33, 39, 50, 68, 69, 76, 80, 85, 86, 95] of visualization a good idea at all? How should we go about collaborating with experts from other domains? What are pitfalls to avoid? How and when should we write a design study paper? These questions motivated and guided our methodological work and we present a set of answers in this paper.

We conducted an extensive literature review in the fields of human computer interaction (HCI) [7, 8, 9, 12, 16, 19, 20, 21, 22, 25, 26, 27, 28, 29, 30, 31, 38, 47, 57, 63, 64, 65, 83] and social science [6, 14, 18, 24, 32, 62, 81, 87, 93] in hopes of finding methodologies that we could apply directly to design study research. Instead, we found an intellectual territory full of quagmires where the very issues we ourselves struggled with were active subjects of nuanced debate. We did not find any off-the-shelf answers that we consider suitable for wholesale assimilation; after careful gleaning we have synthesized a framing of how the concerns of visualization design studies both align

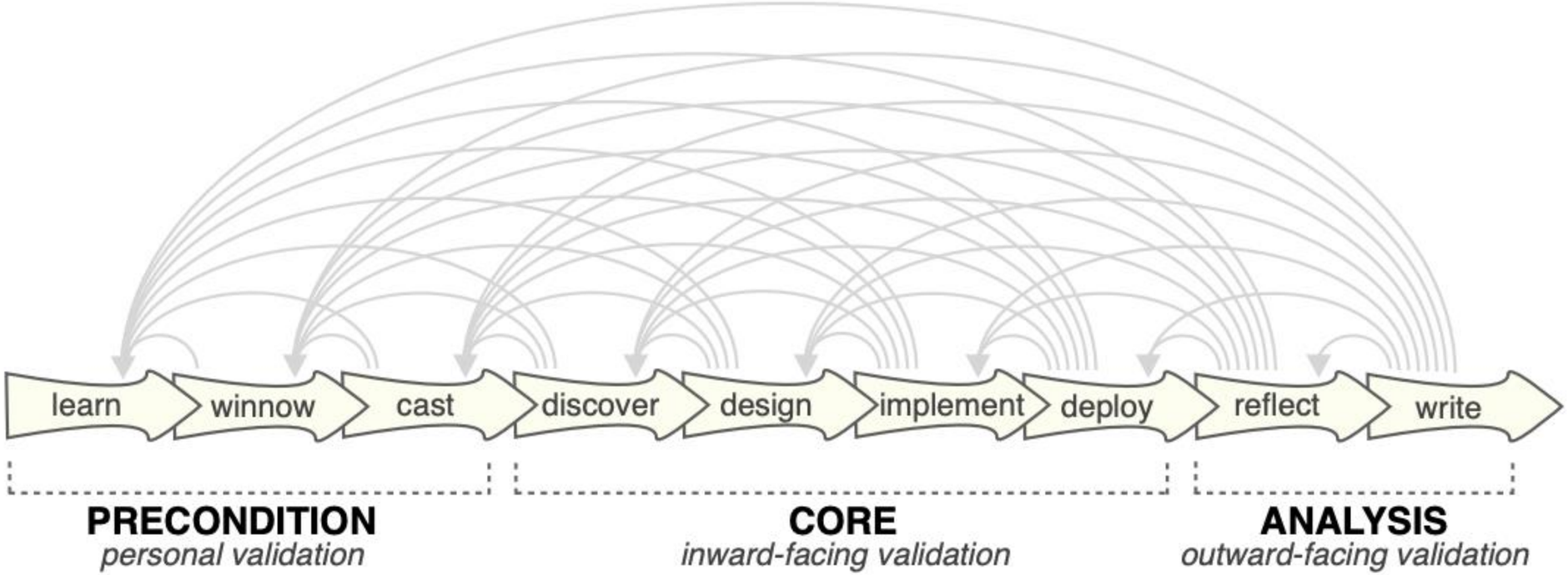


Fig. 2. Nine-stage design study methodology framework classified into three top-level categories. While outlined as a linear process, the overlapping stages and gray arrows imply the iterative dynamics of this process.

Considerations

- Practical
  - Data: Does data exist, is it enough, can you get it?
  - Engagement: How much time do they and you have for the project? How much time can you spend in their environment?
- Intellectual
  - Problem: Is there a vis research question lurking?
  - Need: Is there a real need or are existing approaches good enough?
  - Task: Are you addressing a real task? How long will need persist? How many people care?
- Interpersonal
  - What is your rapport with clients?

When should you use this?

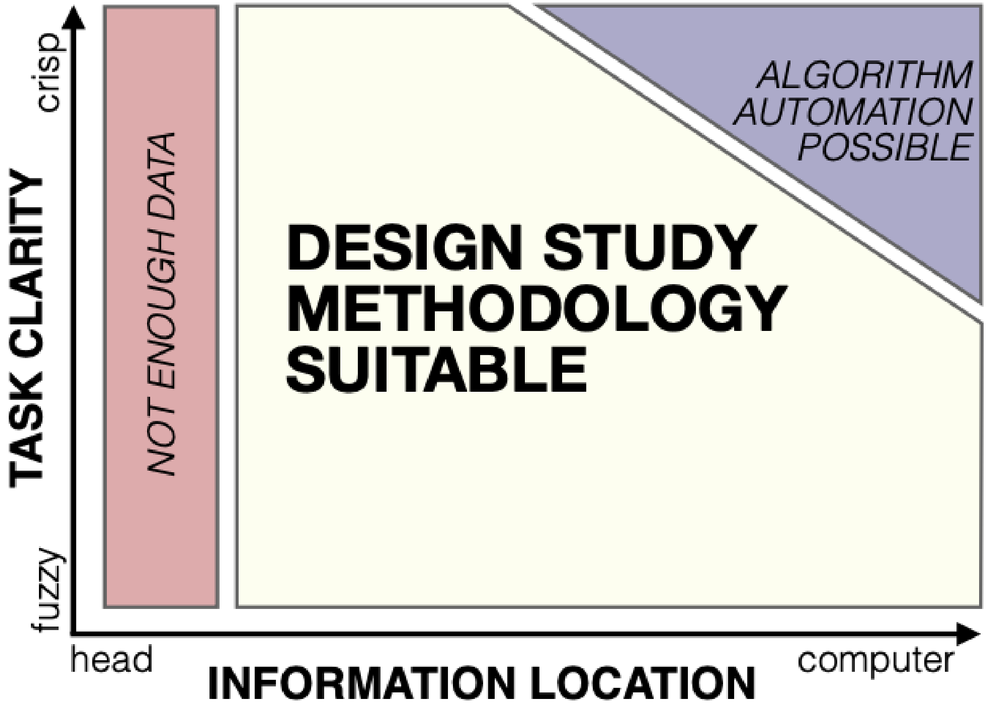


Fig. 1. The task clarity and information location axes as a way to analyze the suitability of design study methodology. Red and blue areas mark regions where design studies may be the wrong methodological choice.

- 32 pitfalls to design study projects listed, organized by framework phase

- Examples
  - No real data available
  - No need for vis, problem can be automated
  - Nonrapid prototyping
  - Premature and insufficient deployment

PF-1	premature advance: jumping forward over stages	general
PF-2	premature start: insufficient knowledge of vis literature	learn
PF-3	premature commitment: collaboration with wrong people	winnow
PF-4	no real data available (yet)	winnow
PF-5	insufficient time available from potential collaborators	winnow
PF-6	no need for visualization: problem can be automated	winnow
PF-7	researcher expertise does not match domain problem	winnow
PF-8	no need for research: engineering vs. research project	winnow
PF-9	no need for change: existing tools are good enough	winnow
PF-10	no real/important/recurring task	winnow
PF-11	no rapport with collaborators	winnow
PF-12	not identifying front line analyst and gatekeeper before start	cast
PF-13	assuming every project will have the same role distribution	cast
PF-14	mistaking fellow tool builders for real end users	cast
PF-15	ignoring practices that currently work well	discover
PF-16	expecting *just talking* or *fly on wall* to work	discover
PF-17	experts focusing on visualization design vs. domain problem	discover
PF-18	learning their problems/language: too little / too much	discover
PF-19	abstraction: too little	design
PF-20	premature design commitment: consideration space too small	design
PF-21	mistaking technique-driven for problem-driven work	design
PF-22	non-rapid prototyping	implement
PF-23	usability: too little / too much	implement
PF-24	premature end: insufficient deploy time built into schedule	deploy
PF-25	usage scenario not case study: non-real task/data/user	deploy
PF-26	*liking* necessary but not sufficient for validation	deploy
PF-27	failing to improve guidelines: confirm, refine, reject, propose	reflect
PF-28	insufficient writing time built into schedule	write
PF-29	no technique contribution $\neq$ good design study	write
PF-30	too much domain background in paper	write
PF-31	story told chronologically vs. focus on final results	write
PF-32	premature end: win race vs. practice music for debut	write

Example

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SafetyLens: Visual Data Analysis of Functional Safety of Vehicles

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Alex Endert‡
Georgia Institute of Technology

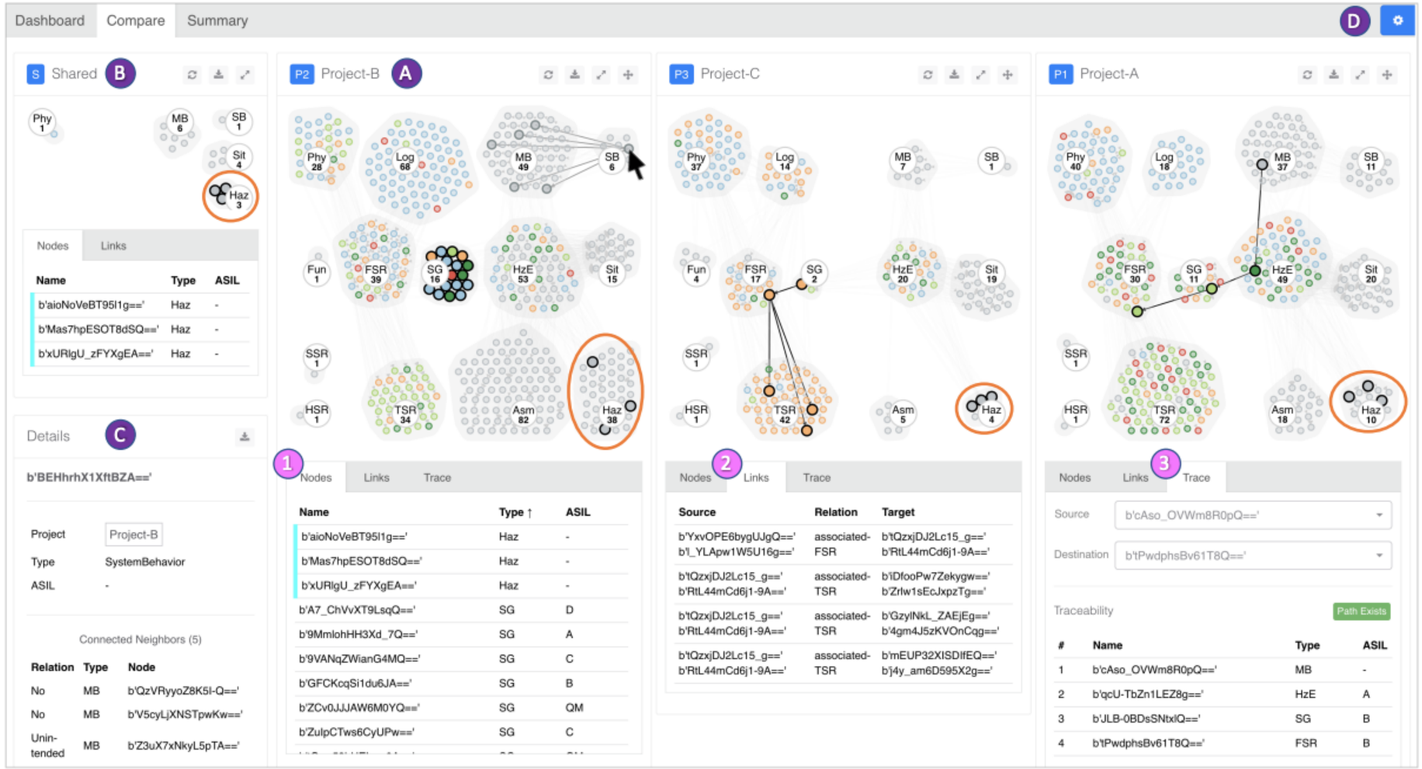


Fig. 1. The SafetyLens Compare view: **(A)** Project Panel comprising the visualization canvas and three tabs: **(1)** Nodes, **(2)** Links, and **(3)** Trace, **(B)** Shared Nodes Panel with three nodes (*Hazards*) selected and highlighted across all three projects (ellipses), **(C)** Detail View Panel showing details of a node (*System Behavior*) hovered in *Project-B* (↖), and **(D)** opens into a Control Panel comprising search and filter controls.

Abstract— Modern automobiles have evolved from just being mechanical machines to having full-fledged electronics systems that enhance vehicle dynamics and driver experience. However, these complex hardware and software systems, if not properly designed, can experience failures that can compromise the safety of the vehicle, its occupants, and the surrounding environment. For example, a system to activate the brakes to avoid a collision saves lives when it functions properly, but could lead to tragic outcomes if the brakes were applied in a way that's inconsistent with the design. Broadly speaking, the analysis performed to minimize such risks falls into a systems engineering domain called Functional Safety. In this paper, we present SafetyLens, a visual data analysis tool to assist engineers and analysts in analyzing automotive Functional Safety datasets. SafetyLens combines techniques including network

First, we had to learn about about the domain

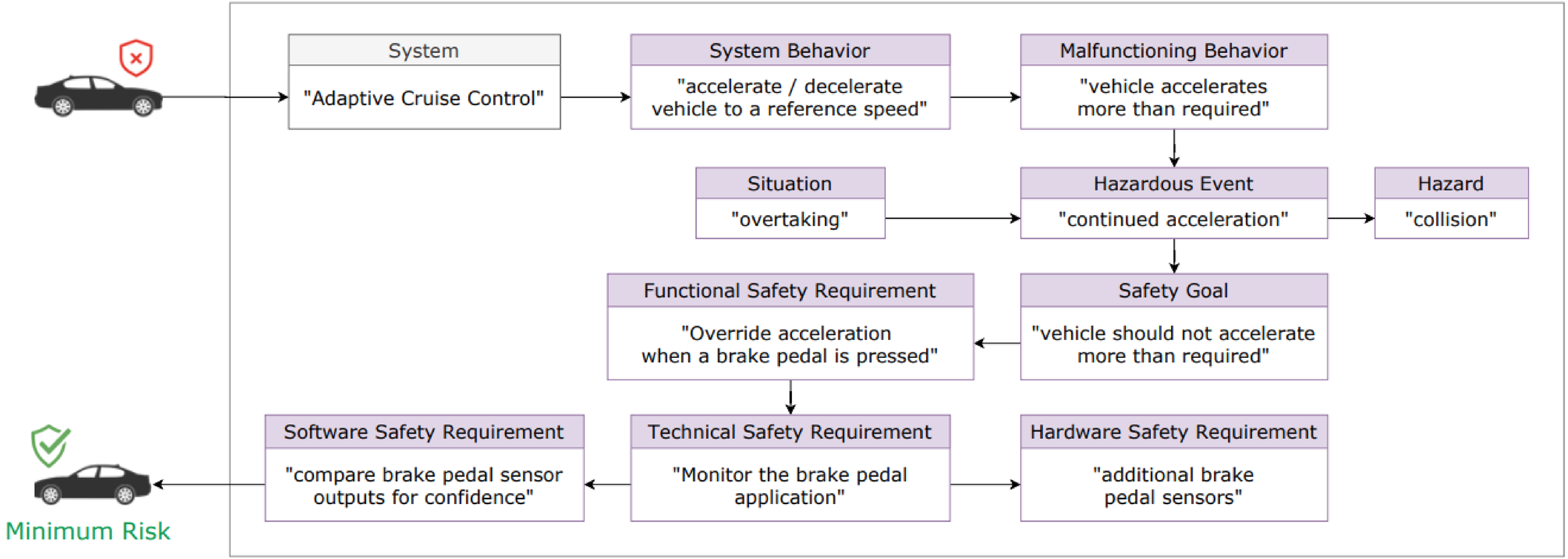


Fig. 2. A hypothetical scenario of an *"Adaptive Cruise Control"* system on which Functional Safety is performed. Purple boxes indicate the *Element Types* described in Table 1

First, we had to learn about about the domain

Element Type	Abb.	Description
System Behavior	SB	An expected behavior of an electrical/electronic *System* in a vehicle.
Malfunctioning Behavior	MB	A failure or unintended behaviour of an item with respect to its design intent.
Situation	Sit	A vehicular operational scenario.
Hazard	Haz	A potential source of harm.
Hazardous Event	HzE	A dangerous condition arising out of a *Situation* and a *Hazard*.
Safety Goal	SG	A requirement to eliminate (or minimize) *Hazards*.
Functional Safety Requirement	FSR	A requirement to achieve the functional or behavioral aspects of a *Safety Goal*.
Technical Safety Requirement	TSR	A technical requirement to meet a *Functional Safety Requirement*.
Software Safety Requirement	SSR	A software-level requirement.
Hardware Safety Requirement	HSR	A hardware-level requirement.

Table 1. A description of the *Types* of *Elements* in the context of automotive Functional Safety. Other Element *Types* such as *Physical*, *Logical*, and *Assumptions* exist but are beyond the scope of this paper.

And also the users, workflow … and create design goals

4.1 User Tasks

User tasks which SafetyLens should support are:

- Discover patterns within projects; e.g., a project has elements that are mostly assigned an ASIL=D (high risk).
- Find missing (otherwise *must-have*) links between nodes; e.g., $\{MB \rightarrow HzE\}$, $\{HzE \rightarrow SG\}$, $\{SG \rightarrow FSR\}$, $\{FSR \rightarrow TSR\}$.
- Look up and analyze a node's end-to-end traceability (i.e., the extent to which functional safety elements have been defined); e.g. $\{MB \rightarrow HzE \rightarrow SG \rightarrow FSR \rightarrow TSR\}$.
- Compare ASILs by analyzing their decomposition into their *S-E-C* (Severity, Exposure, and Controllability) constituents.
- Find common nodes and links among projects; (e.g., *Hazards (Haz)* that exist in multiple projects).
- Discover anomalies across projects; (e.g., the same *Hazardous Event (HzE)* is assigned a different ASIL across projects).
- Compare key metrics across projects; (e.g., counts and distribution of nodes and links within projects).
- Get an overview of current and past projects to monitor status.

4.2 Design Goals

From the design exercises and requirements gathering activities described above we identified the following design goals.

DG1: Facilitate Exploration of a Project User tasks such as "Find missing links", "Lookup and analyze a node's end-to-end traceability" are network exploration tasks. Hence, we modeled Functional Safety data as a network and derived this design goal to support tasks like fetching node details on hover, finding adjacent nodes, and finding paths from one node to another based on the taxonomies by Lee et al. [34] and Pretorius et al. [40].

DG2: Facilitate Comparison among Projects Teams within functional safety can be system-specific and may not always

Finally, we designed and built a prototype

We showed the prototype to domain experts

- Received lots of feedback (not all positive, but all helpful)

How long did it take…

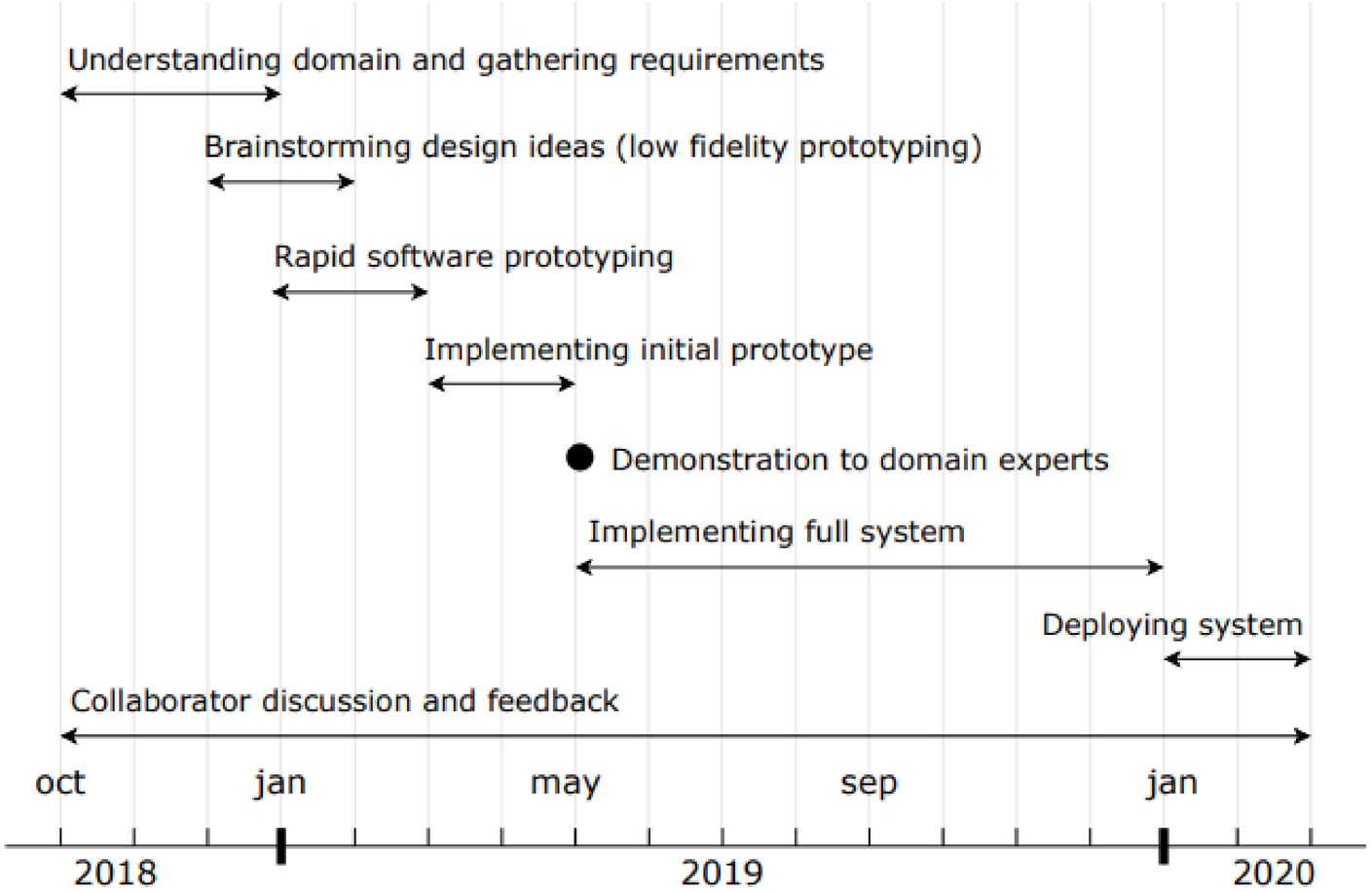
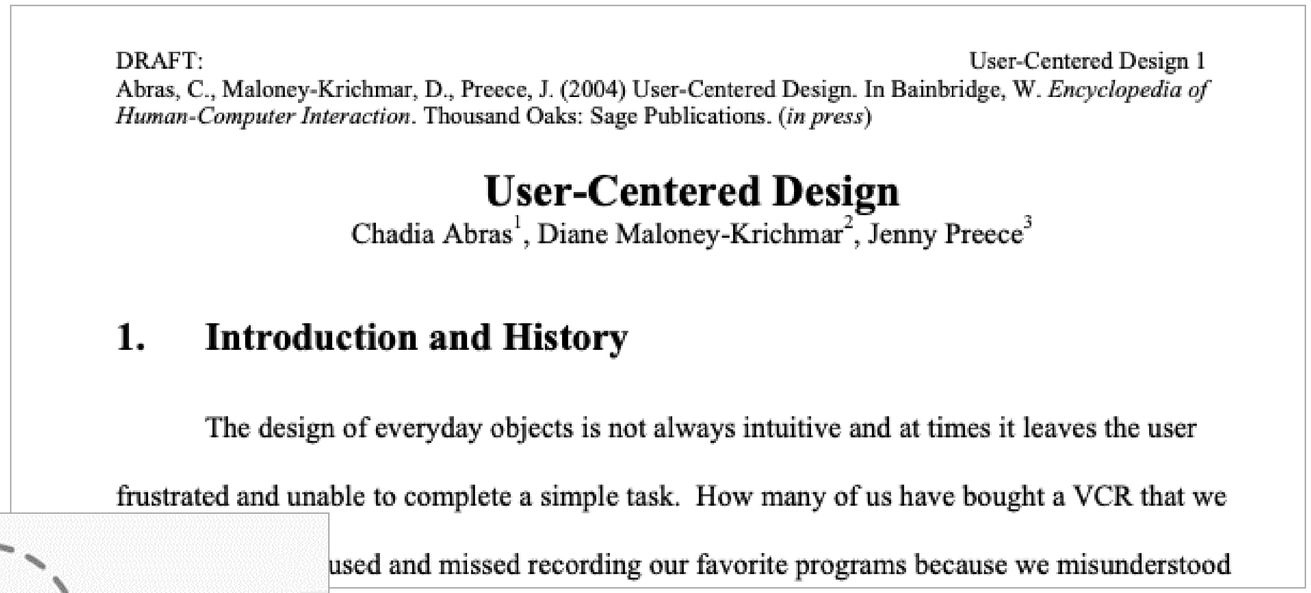


Fig. 3. The high-level phases and milestones of the design process.

Connections to other disciplines

- User-centered design



DRAFT: User-Centered Design 1

Abras, C., Maloney-Krichmar, D., Preece, J. (2004) User-Centered Design. In Bainbridge, W. *Encyclopedia of Human-Computer Interaction*. Thousand Oaks: Sage Publications. (*in press*)

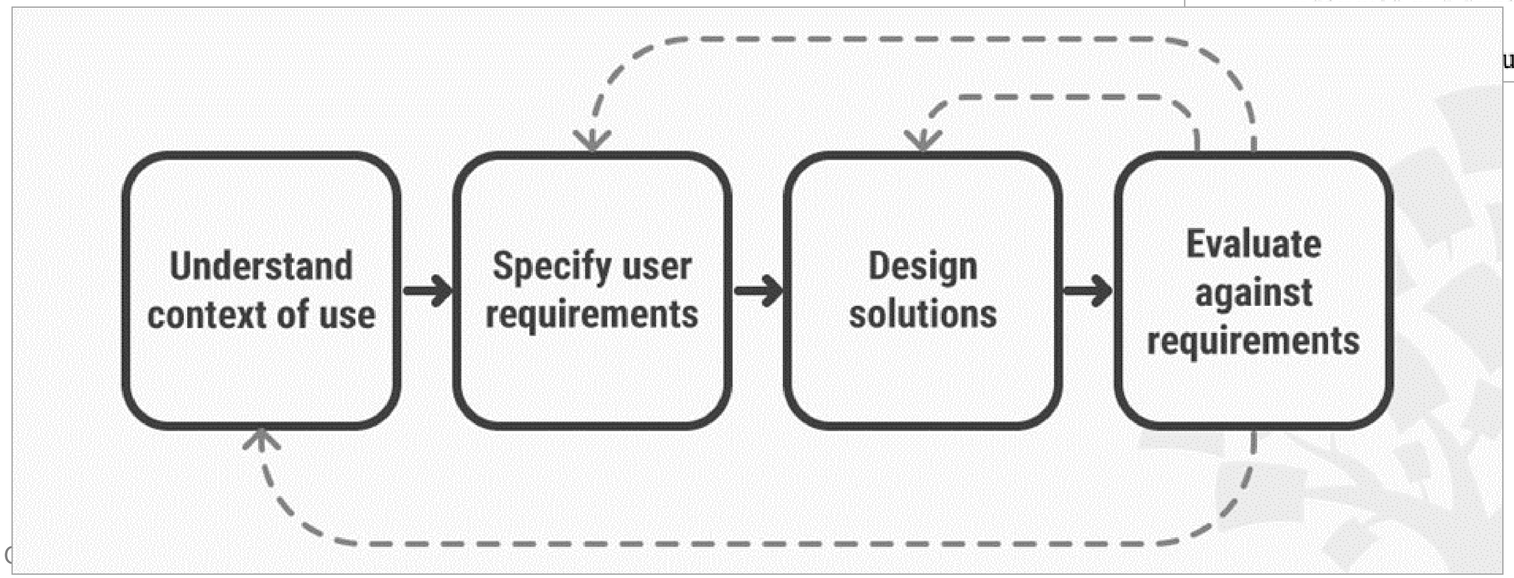
User-Centered Design

Chadia Abras[1], Diane Maloney-Krichmar[2], Jenny Preece[3]

1. Introduction and History

The design of everyday objects is not always intuitive and at times it leaves the user frustrated and unable to complete a simple task. How many of us have bought a VCR that we

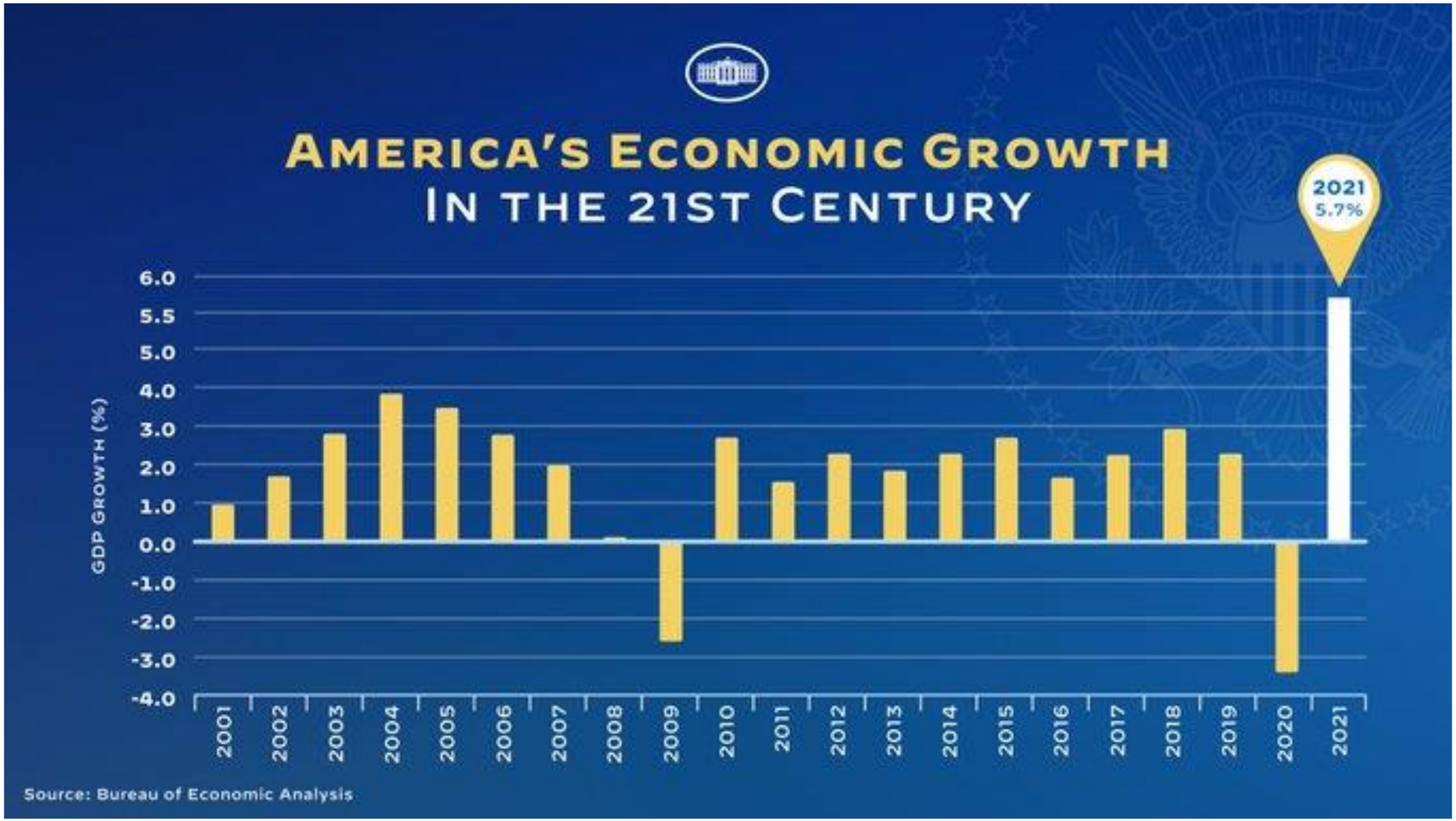
used and missed recording our favorite programs because we misunderstood



Miss these? ;)

Design guidelines and principles – a Preview

What’s wrong/odd?



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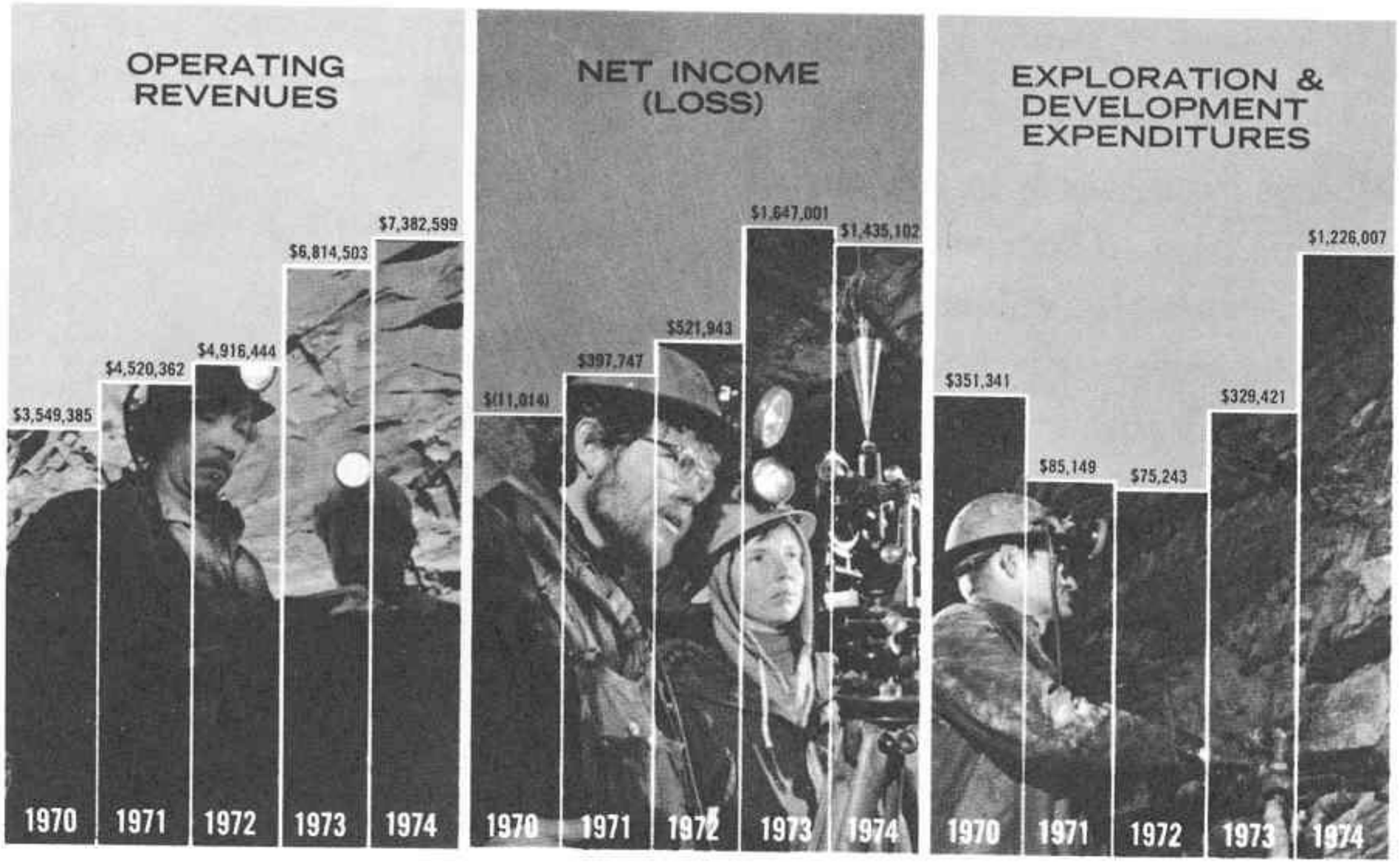

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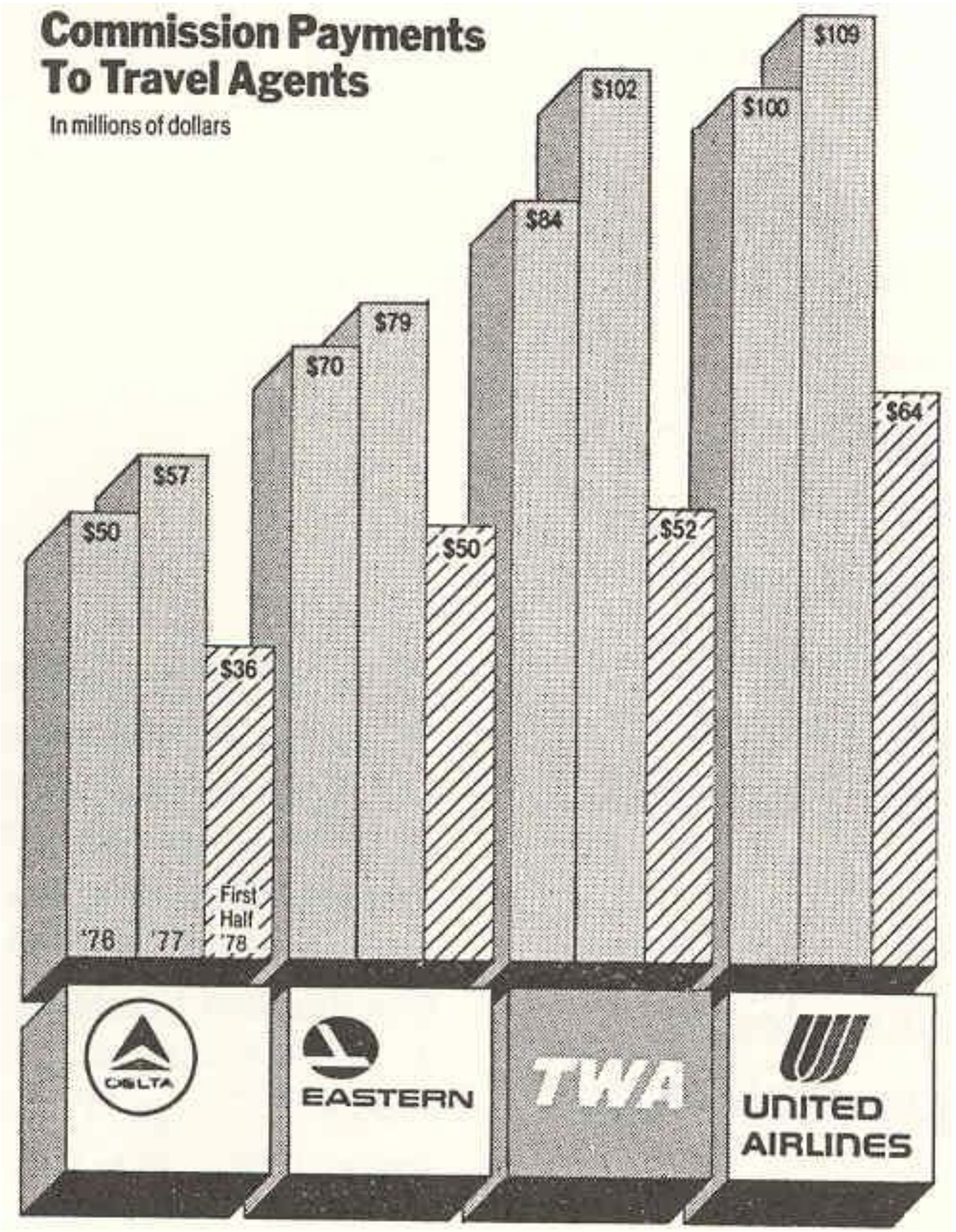

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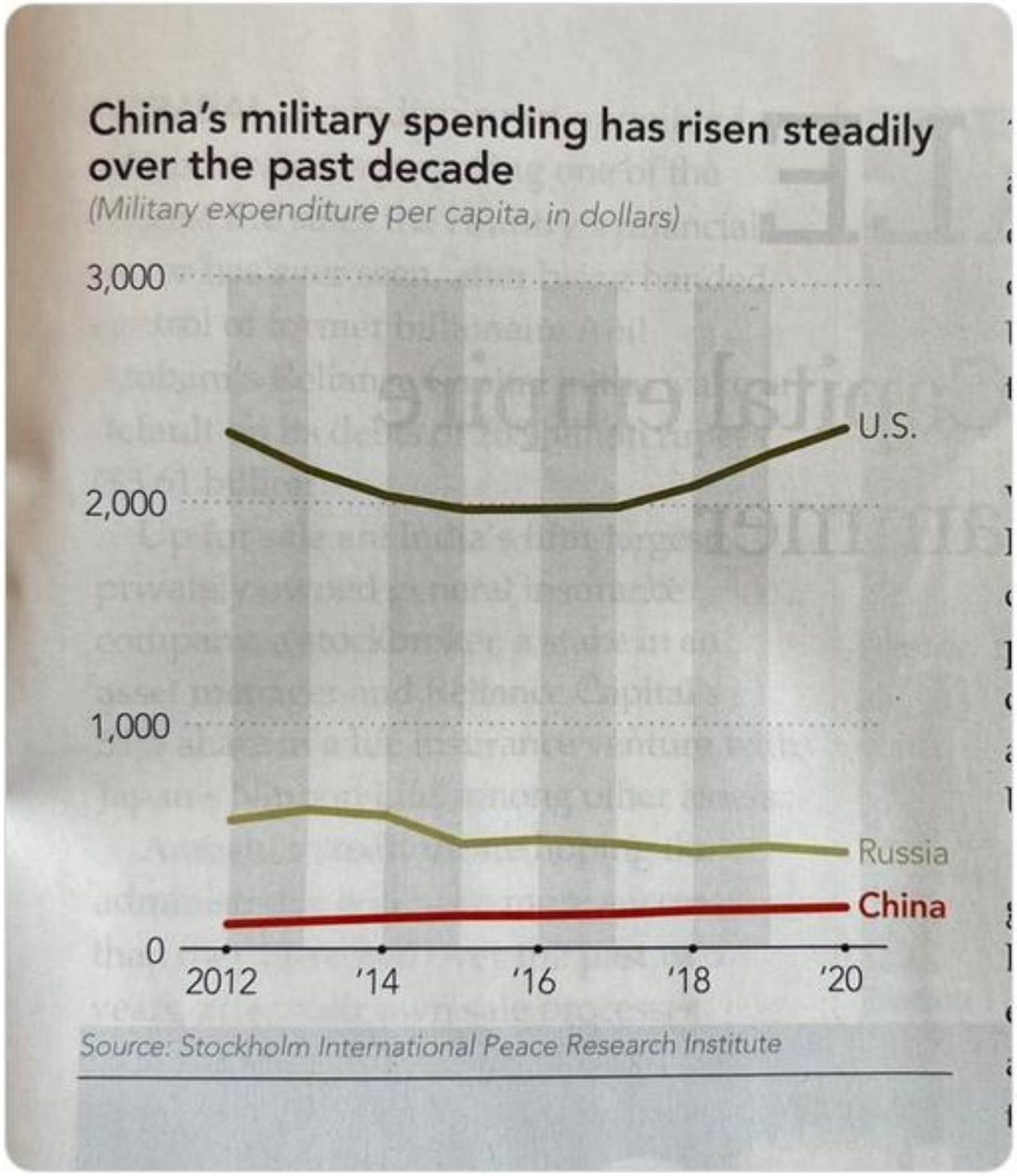

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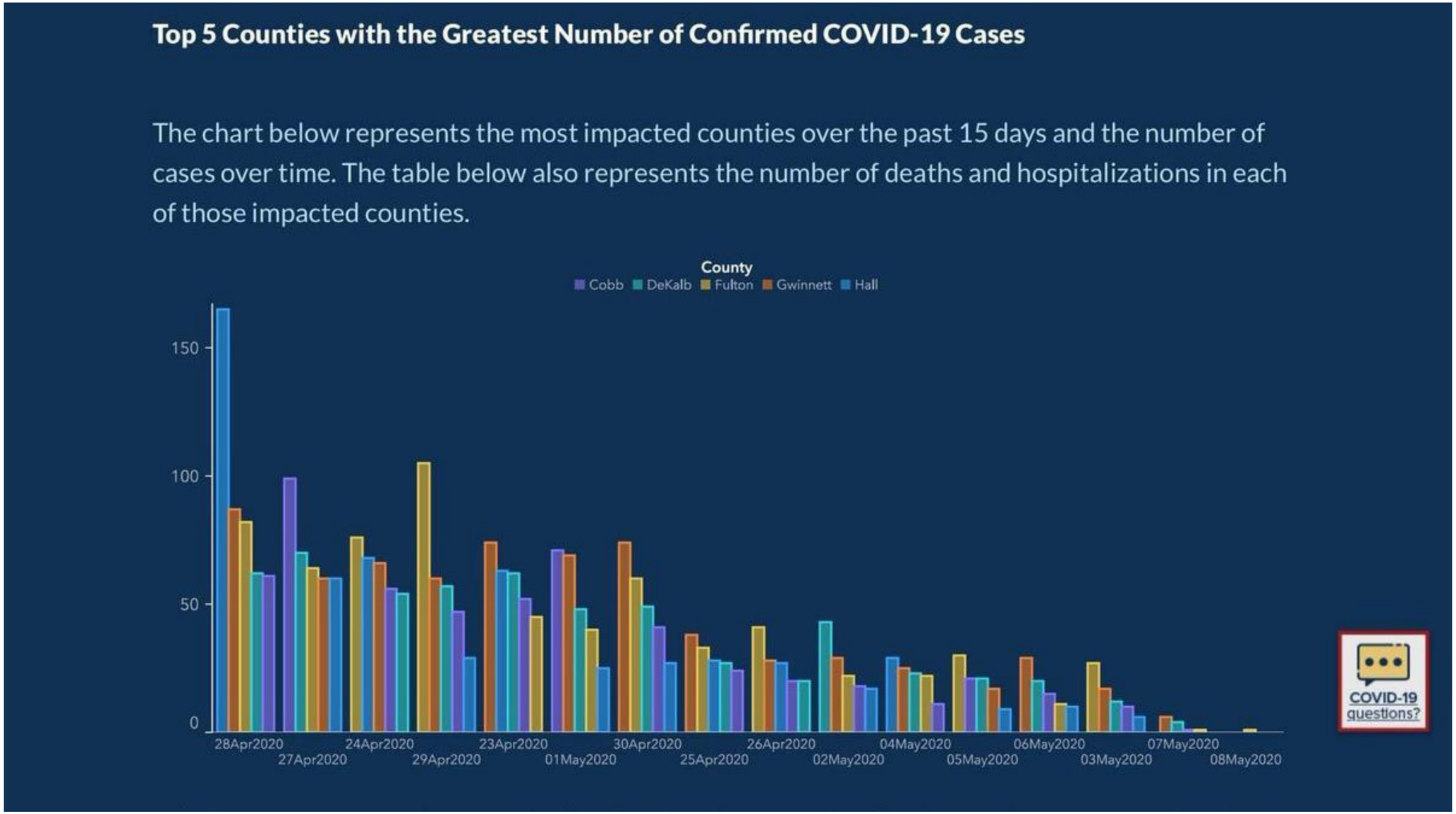

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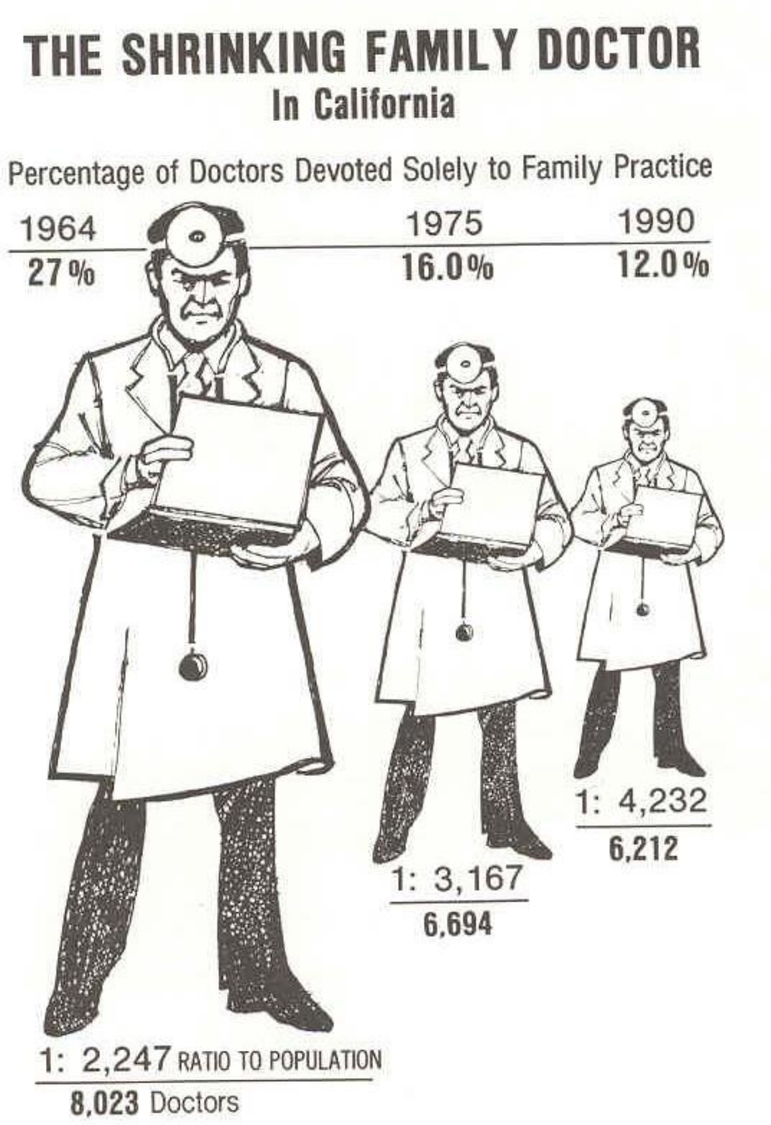
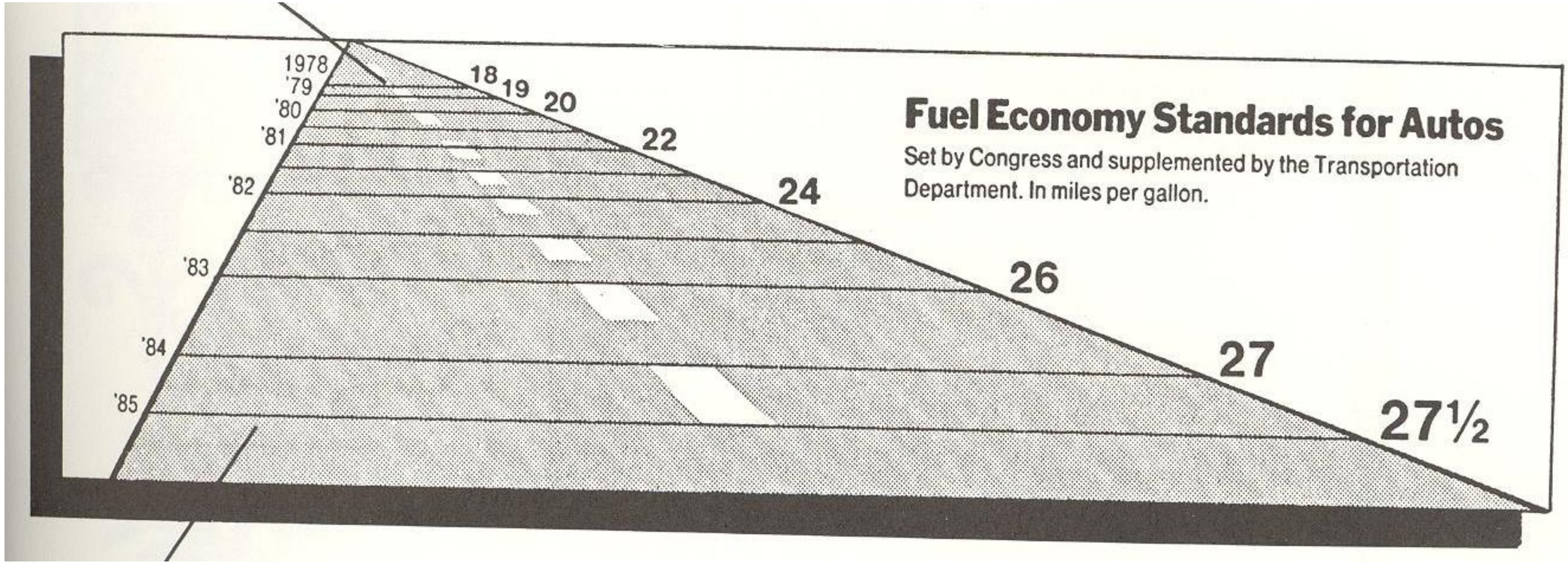


What’s wrong/odd?



What’s wrong/odd?

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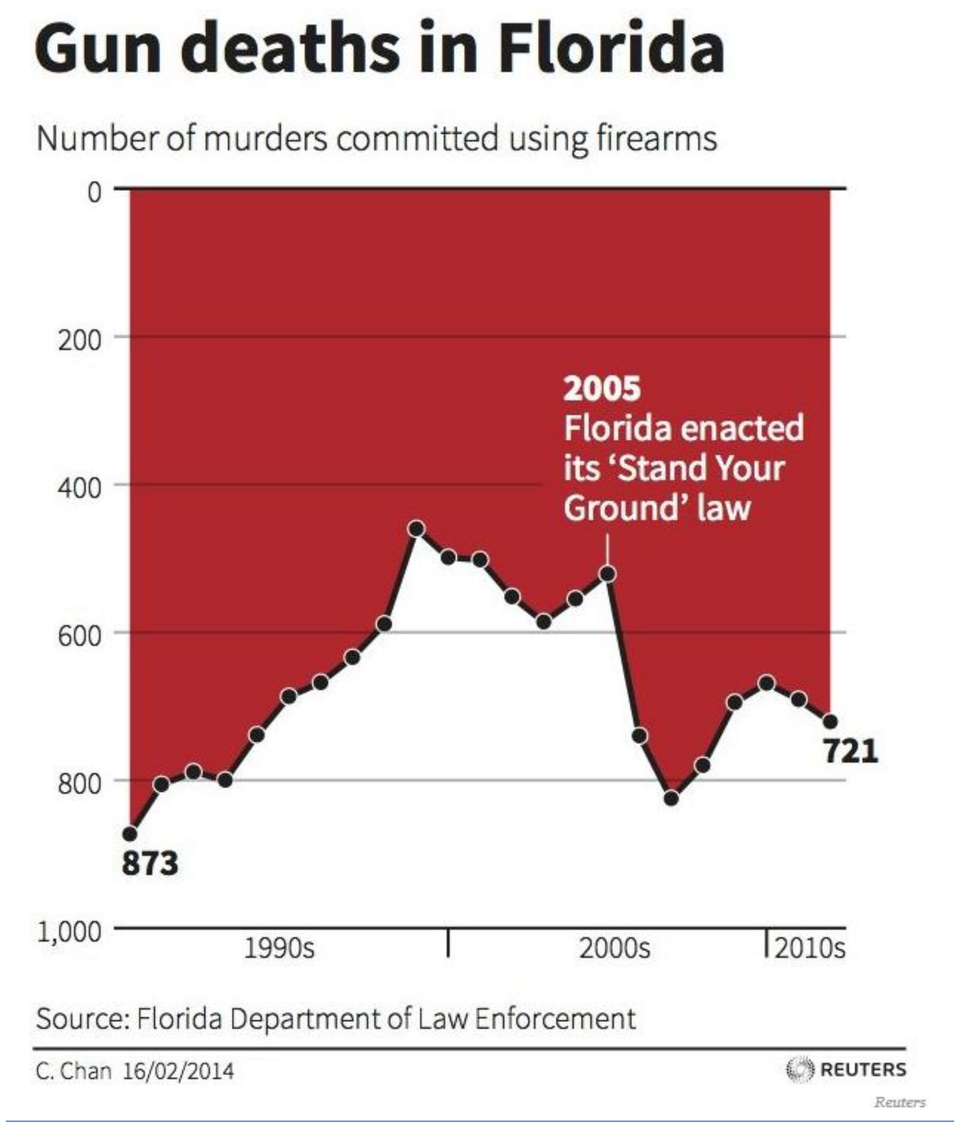
What’s wrong/odd?



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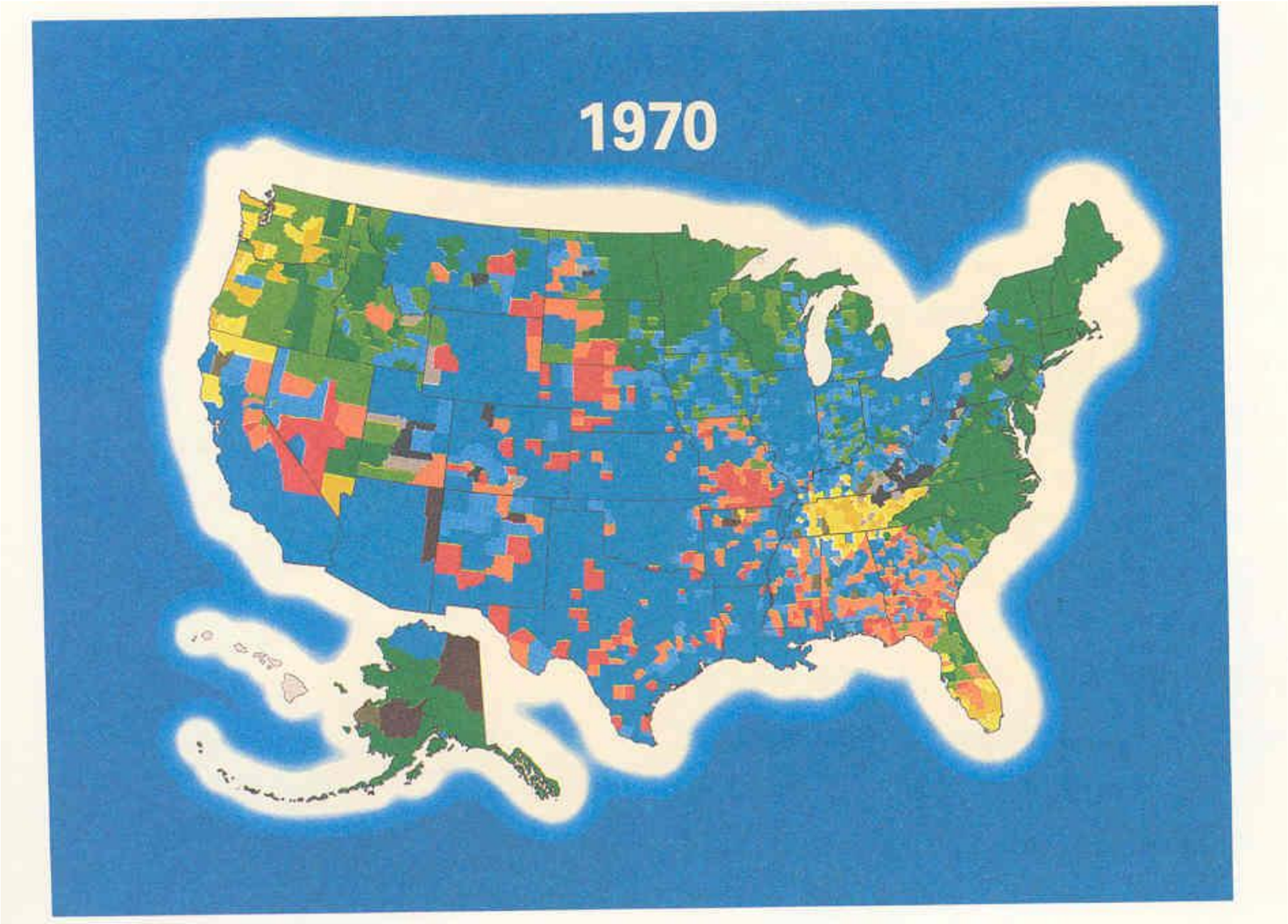

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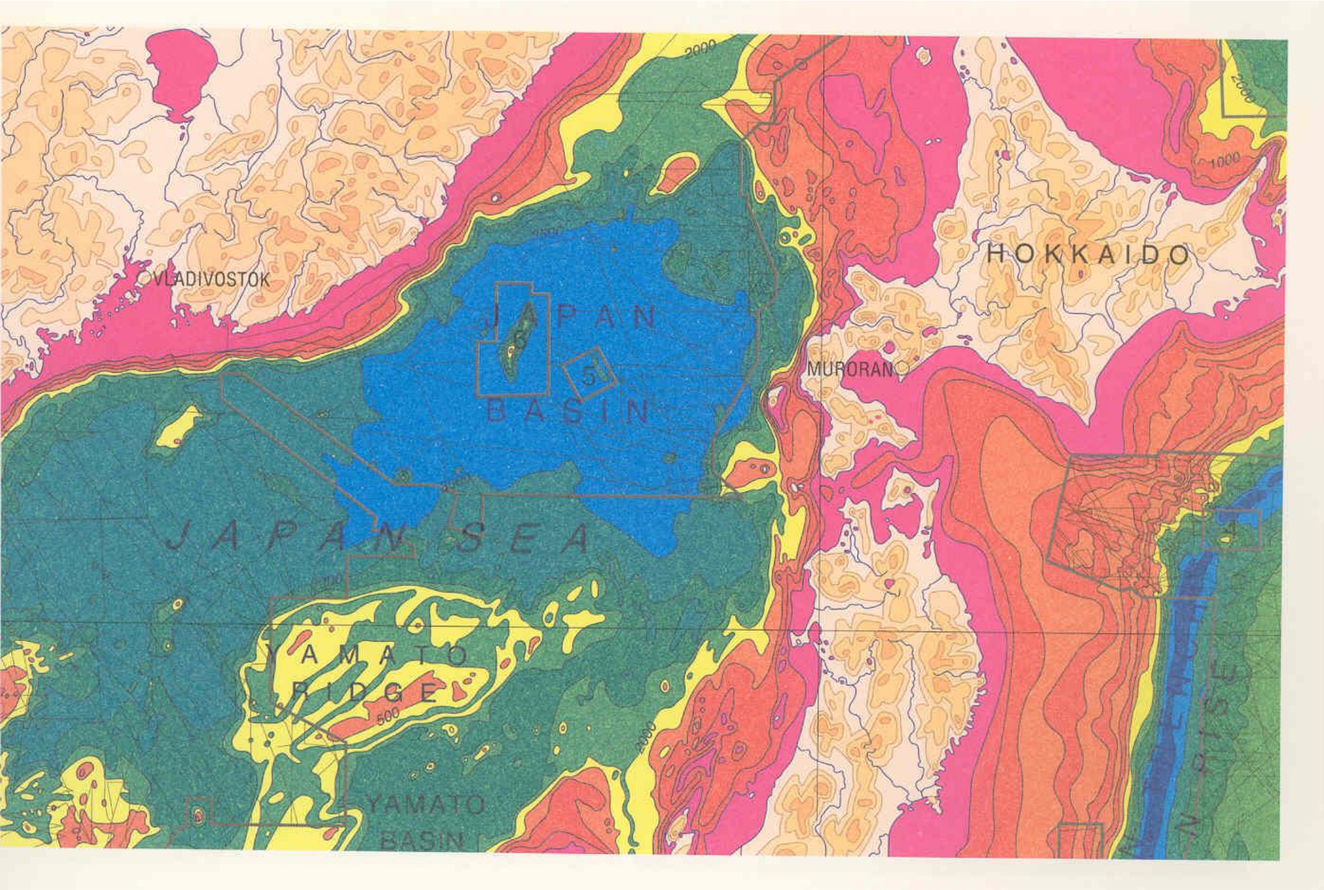

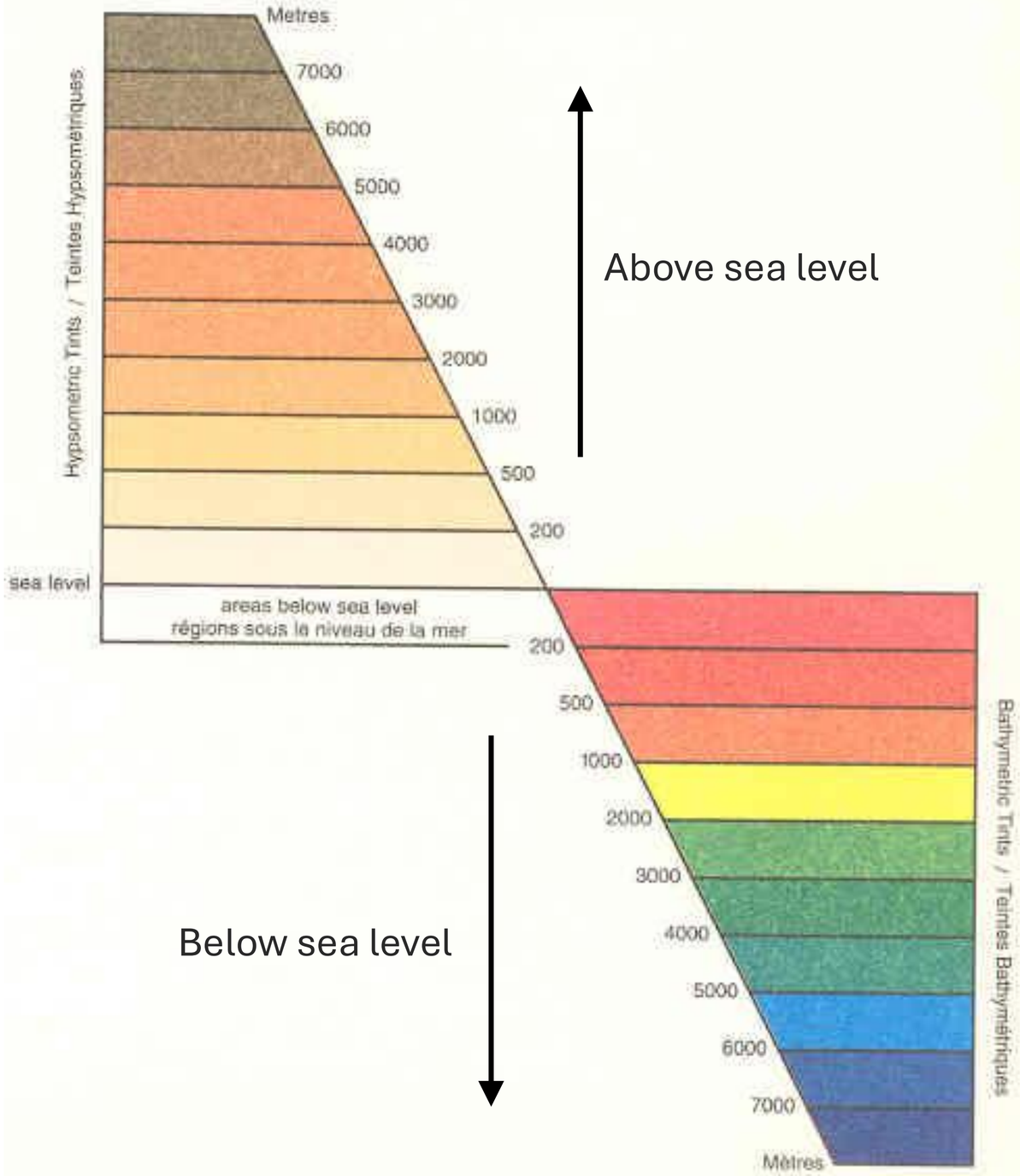

Review

- Design (and redesign) as a process to help us think about the problem we are designing against
- Design studies
- Design guidelines and principles – a preview

Next time

- Graphical Integrity and Design Aesthetics